



PRACTICAL SECTIONS

Microbial Physiology (Fungi)



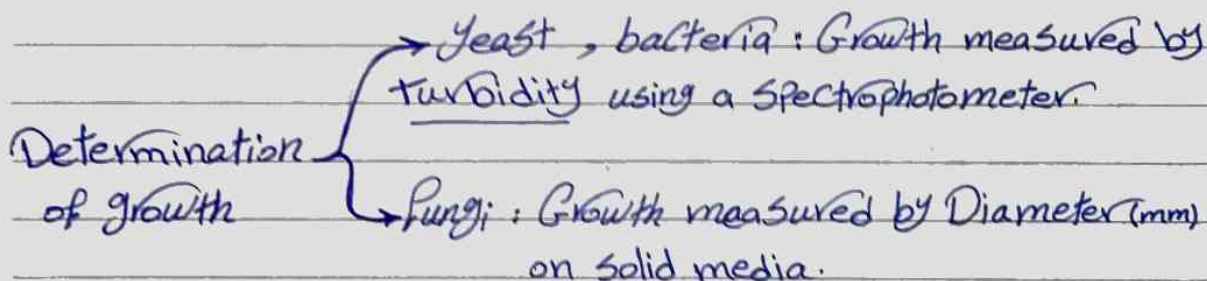
Fungal Physiology

Fungi

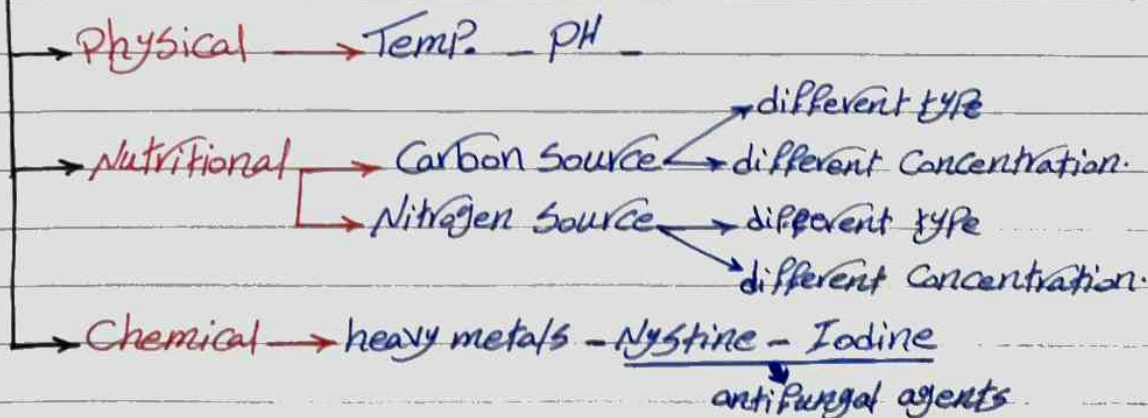
- Eukaryotic → Yeast
- Uni- or multi Cellular
- Heterotrophic, extracellular digestion (exotrophic).
- Acidic (pH 4 → 6.5)
- Temp. 25 → 30°C
- Incubation 3-7 days.

Bacteria

- Prokaryotic.
- unicellular.
- Autotrophic or Heterotrophic
- Neutral to slightly alkaline (pH 7 → 9)
- Temp. 37°C (30 → 40°C)
- Incubation 24-48 hours.



Factors affecting Fungal growth



Physical Factors

Effect Temp. in Fungal Growth.

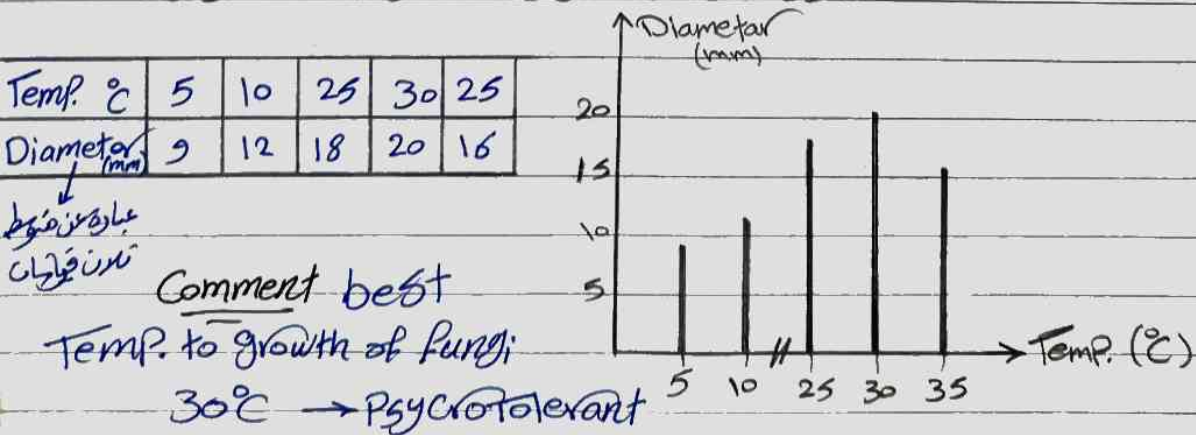
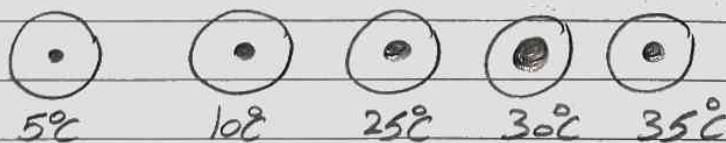
Aim To study the effect of different temperatures on the growth rate of fungi.

Principle Temperature affects enzyme activity and metabolic reactions.
→ Each microorganism has a minimum, optimum, and maximum temperature for growth.

→ For most fungi, the optimum temperature is around 25-30°C.

Procedure

1. Prepare Dox media
2. Sterilization by autoclave then → Pour it into Petri dishes.
3. Inoculation by test organism (*Aspergillus* or *Fusarium*)
4. Incubation for 4 days at different temperatures (5, 10, 25, 30, 35)°C
5. Determine growth by Diameter (mm)

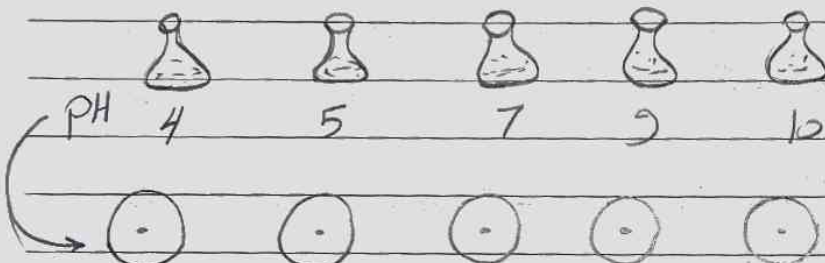


→ Effect of pH

Aim: To determine the influence of different pH values on Fungal growth

Principle → pH affects enzyme activity, nutrient solubility, and Cell membrane function.

→ Most fungi prefer slightly Acidic conditions (pH 4–6.5).

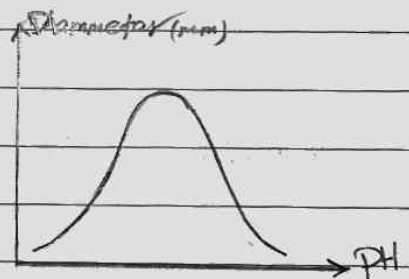


This Fungi is
Alkalotolerant
→ most acidic.

At Temp. 25 → 30°C & incubation for 5 days



Diameter (mm)	—	—	—	—	—
pH	4	5	7	9	10



Hints

<u>pH</u>	<u>type</u>
7	Neutrophilic
4 → 8	Alkalotolerant
6 → 10	Acidotolerant
4 → 6.5	Acidophilic
8 → 14	Alkalophilic

Nutritional Factors

Effect of different Carbon Source.

Aim: To determine the best carbon source on fungal growth.

Principle:

- ① Mono-saccharides (Simple Sugars) are used mainly by most fungi.

- ② Fungal (Microbes) used carbon source for energy production (ATP).

- ③ Different carbon sources affect fungal growth depending on how easily the organism can metabolize them.

Procedure:

- 1- Prepare 5 media plates - each containing a different carbon source:

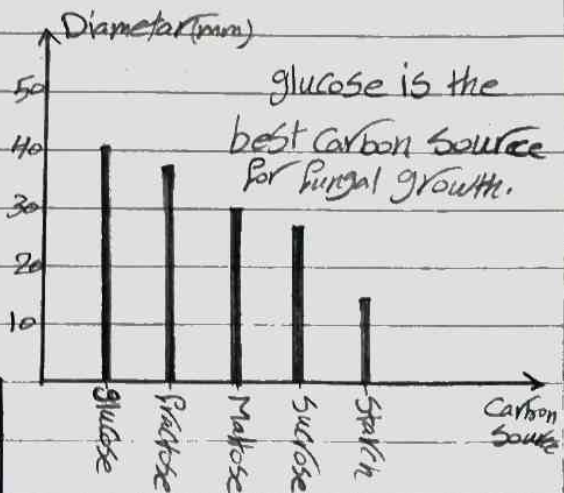


glucose fructose maltose sucrose starch

2- Sterilization in autoclave at 121°C , 1.5 atm, for 20 minutes.



3- Incubation at 30°C for 5 days



Carbon Source	glucose	fructose	maltose	sucrose	starch
Diameter (mm)	40	38	30	28	14

→ What happens if the glucose concentration increases above 20g/L ?
 (Osmotic pressure of the medium increases causing ~~plasmolysis~~ plasmolysis and reduced fungal growth.)

Effect of different Nitrogen Source

Aim: To determine the best ^{Nitrogen} ~~carbon~~ source on fungal growth.

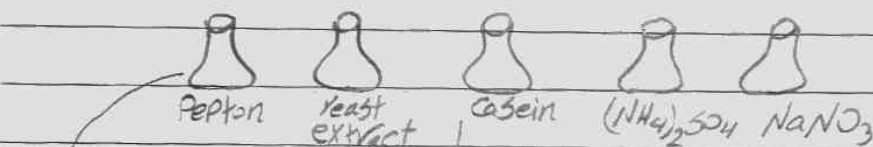
Principle: Two Type of Nitrogen Source

→ Organic: Pepton - Casein - Beef extract - Yeast extract.

→ Inorganic: $(\text{NH}_4)_2\text{SO}_4$ - KNH_4 - NaNO_3 - $(\text{NH}_4)_3\text{PO}_4$

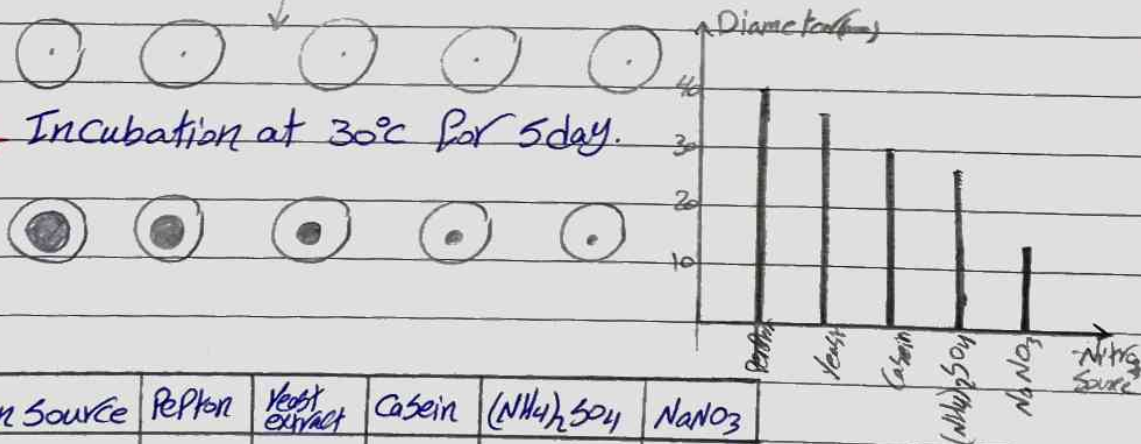
• Organism used Nitrogen Source for growth.

Procedure: 1- Prepare media (5 plates) - each containing a different Nitrogen Source.



2- Sterilization in autoclave at 121°C , 1.5 atm, for 20 min.

3- Incubation at 30°C for 5 day.



Nitrogen Source	Pepton	Yeast extract	Casein	$(\text{NH}_4)_2\text{SO}_4$	NaNO_3
Diameter (mm)	40	38	30	28	14

Peptone is the best Nitrogen Source for fungal growth.